# The Cool Ratio Lemma

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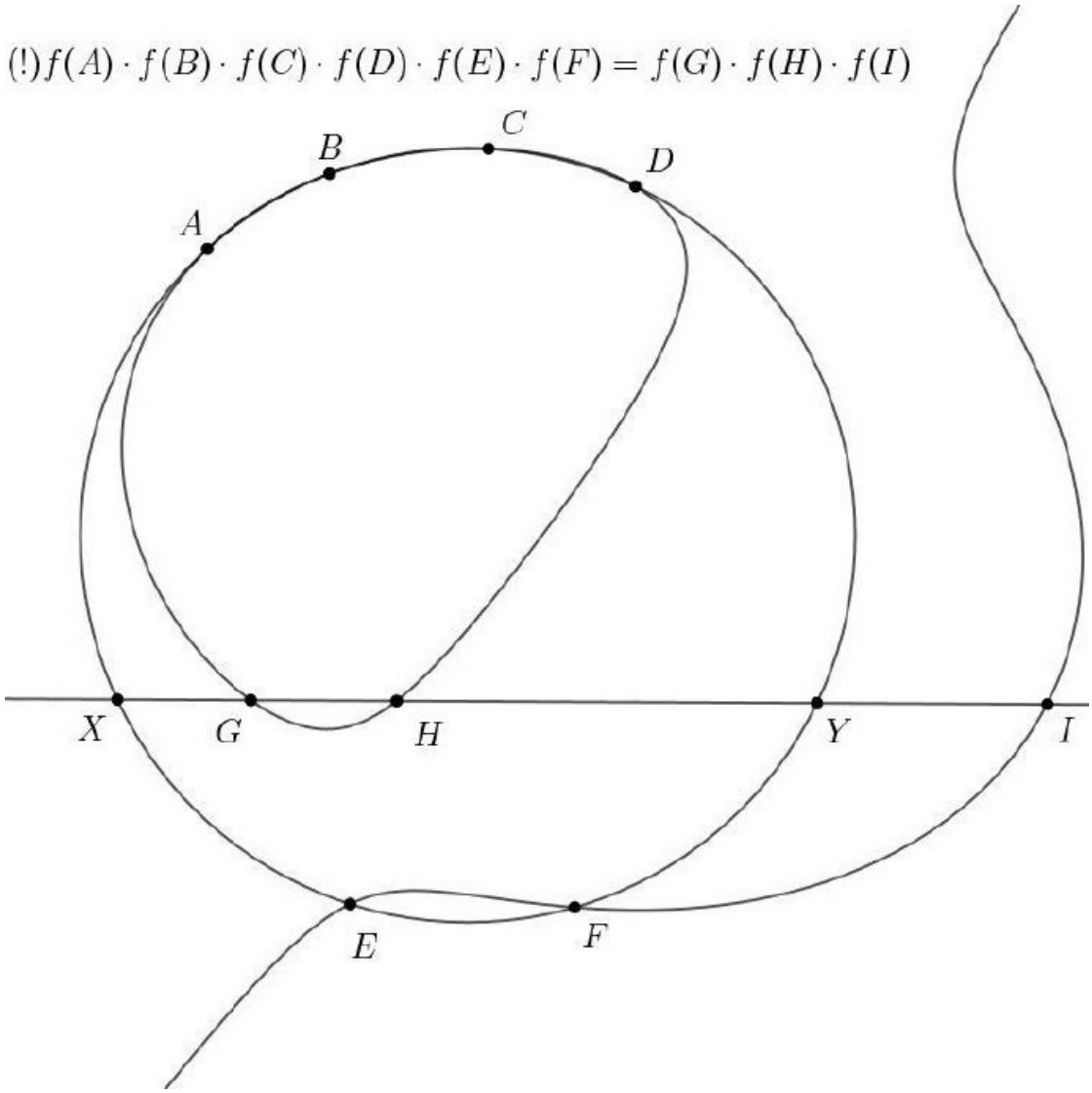
*"CRL is like a swiss knife"* – whatisthis, 3rd AGO winner



# §1 Lecture notes

## §1.1 Introduction

In synthetic geometry, it is not enough to master angles; one must also master ratios between segments. Although this is not a problem with trigonometry, resorting to it can lead to long and complicated calculations. The *cool ratio lemma*, a result formalized by the author mira74[2], is a particularly strong result because it allows for the extraction of information about relative lengths between certain points without heavy computations. Several authors have discussed this lemma [2, 1] and this handout is just a compilation of facts from both handouts, most credits go to them.

## §1.2 Motivation

Let's start by citing a very famous theorem regarding ratios.

**Theorem 1.1** (Generalized Angle Bisector Theorem)
Let $ABC$ be a triangle, where $D$ is a point on the line $BC$. Then,

$$\frac{BD}{CD} = \frac{BA}{CA} \cdot \frac{\sin \angle BAD}{\sin \angle CAD}$$

While this theorem allows us to extract information about segments in a triangle, we can make it even stronger when we bring the circumcircle into play. With this in mind, we can prove the following result.

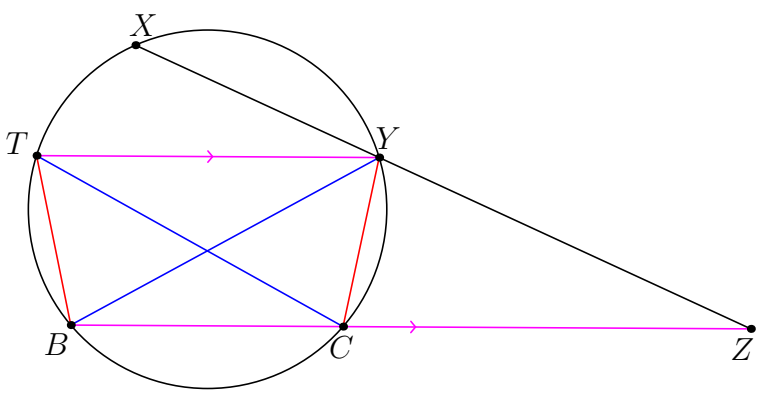
**Theorem 1.2** (Cool Ratio Lemma)
Let $\omega$ be a circle passing through points $B$ and $C$. Consider a line that intersects $\omega$ at points $X, Y$ and $BC$ at $Z$. Then

$$\frac{BZ}{CZ} = \frac{BX}{CX} \cdot \frac{BY}{CY}$$

*Proof.* Although we can show this with trigonometry, I present a more intuitive solution.

Let $T$ be the intersection of the line parallel to $BC$ that passes through $Y$. Then we have:

$$\frac{BZ}{CZ} = (BC; Z\infty_{BC}) = (BC; XT) = \frac{BX \cdot CT}{CX \cdot BT} = \frac{BX}{CX} \cdot \frac{BY}{CY}$$




□

To simplify notation, we can write the ratios in terms of a function.

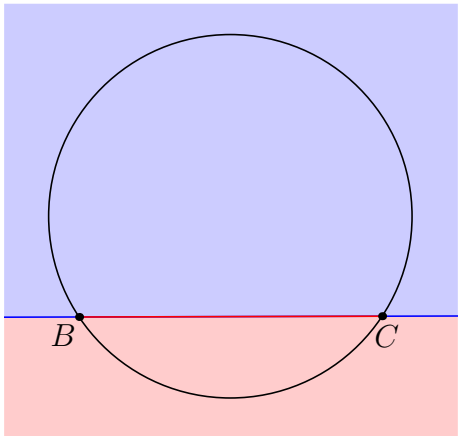
**Definition 1.3.** Let $B, C$ be points in the plane and $f_{BC} : \mathbb{R}^2 \backslash C \to \mathbb{R}$, where

$$f_{BC}(P) = \pm \frac{BP}{CP}$$

is negative when $P$ is on segment $BC$ or below it, and positive otherwise. We call this the *cool ratio* function over $BC$.

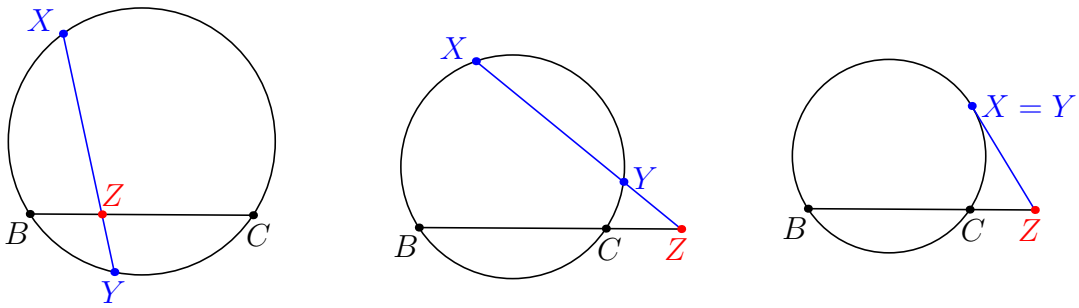
**Remark 1.4.** Often, we only define the cool-ratio function for a single segment. To simplify writing, we use $f = f_{BC}$.

**Remark 1.5.** It is important to note that $f_{BC} \neq f_{CB}$.




Therefore, the CRL reduces to the following equation:

$$f(Z) = f(X)f(Y).$$




The sign assignment will be useful for describing the properties of function $f$. In fact, if we consider a circle $\omega$ with a chord $XY$ on which we define the function $f_{XY}$, we can conclude that $f_{XY}$ is a bijection from $\omega \backslash Y$ to $\mathbb{R}$.

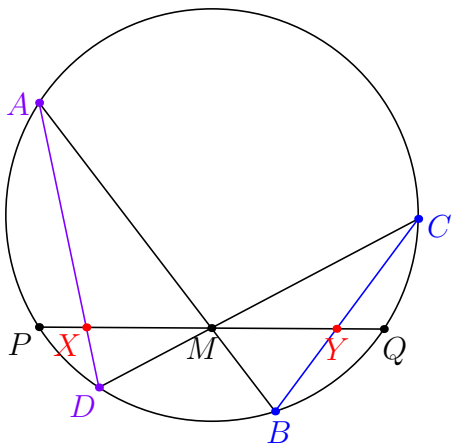
**Lemma 1.6** (Properties of $f$)

Let $B, C$ be points in the plane and $f(P) = \pm\frac{BP}{CP}$. Then,

- Given points $X, Y$ on the same circle or line passing through $B$ and $C$, we have
$$f(X) = f(Y) \Leftrightarrow X = Y.$$
- If $BCXY$ is cyclic and $BC$ intersects $XY$ at $Z$, then $f(X)f(Y) = f(Z)$.

**Example 1.7** (Butterfly Theorem)

Let $M$ be the midpoint of a chord $PQ$ of a circle, through which two other chords $AB$ and $CD$ are drawn; $AD$ and $BC$ intersect the chord $PQ$ at $X$ and $Y$, respectively. Then, $M$ is the midpoint of $XY$.




**Walkthrough.** The primary step is to figure out which points $U$ and $V$ to choose to define the function. Usually, this choice is motivated by the number of lines and circles passing through $UV$.

Next, calculate as many terms as possible in terms of other points.

1. Let $f(Z) = \pm\frac{PZ}{QZ}$.
2. Calculate $f(X)$ and $f(Y)$.
3. Prove that $f(X)f(Y) = 1$.

### §1.2.1 Why so much hate on CRL?

There is some resistance to using the CRL, which usually arises from how it is presented: equalities between segments that don't seem natural and a procedure that sounds overly algebraic. This first impression leads many to see it as something distant from synthetic geometry.

However, this criticism loses strength when we compare the CRL to well-established tools. *Cross-ratios*, for example, are also defined using segments and sines, but are rarely questioned from this perspective. They are accepted because they are not treated as mere numerical expressions, but rather as fundamental invariants of projections and projective transformations, capable of generalizing across different geometric contexts.

Thus, when we use cross-ratios, we operate with their essential properties rather than the concrete elements in their definition. The same reasoning can be applied to CRL: instead of seeing it as a calculation between segments, it is more productive to view it as a structural object that is preserved and transported throughout the geometric problem.

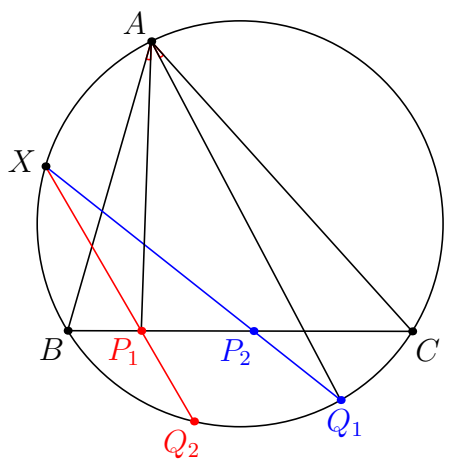
### §1.2.2 More examples!

**Example 1.8**

Let $\triangle ABC$ be a triangle, and let $P_1$ and $Q_1$ be points on $BC$ and on the circumcircle of $\triangle ABC$, respectively, such that

$$\angle BAP_1 = \angle CAQ_1.$$

Points $P_2$ and $Q_2$ are defined analogously. Prove that the lines $P_1Q_2$ and $P_2Q_1$ intersect on the circumcircle of $\triangle ABC$.



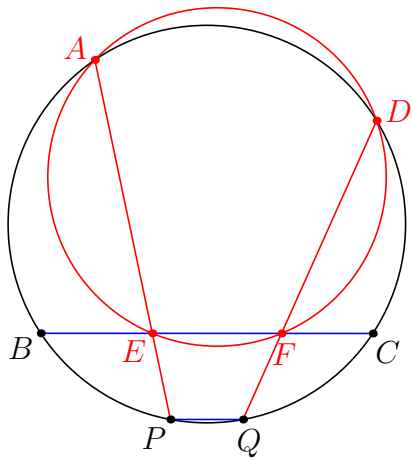

**Walkthrough.** Notice that by Lemma 1.4, if we define points $X_1 = P_1Q_2 \cap (ABC)$ and $X_2 = P_2Q_2 \cap (ABC)$, to prove that $X_1 = X_2$ it is sufficient to show $f(X_1) = f(X_2)$.

1. Let $f(Z) = \pm\frac{BZ}{CZ}$.
2. Use the fact that $\angle BAP_1 = \angle CAQ_1$ to calculate $f(P_1)$.
3. Reflect $Q_1$ and $Q_2$ across the perpendicular bisector of $BC$.
4. Prove that $f(X_1) = f(X_2)$.

**Theorem 1.9** (CRL: circle + line)

Let $ABCD$ be a cyclic quadrilateral, and let $E$ and $F$ be points on the line $BC$. Let $f(Z) = \pm\frac{BZ}{CZ}$ with the usual sign definition. Then

$$ADEF \text{ is cyclic} \Leftrightarrow f(A)f(D) = f(E)f(F)$$




*Proof.* Let $P$ and $Q$ be the second intersection of lines $AE$ and $DF$ with $(ABCD)$.

By Reim's theorem, $ADEF$ is cyclic if and only if $PQ \parallel BC$. This is equivalent to

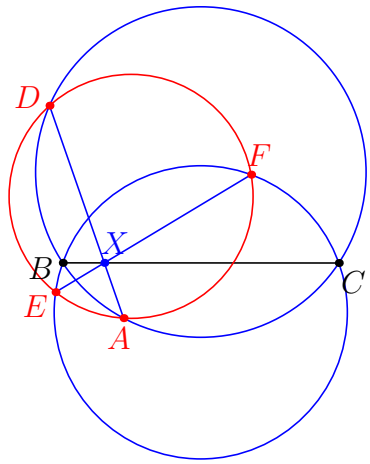
$$1 = f(P)f(Q) = \frac{f(E)}{f(A)} \cdot \frac{f(F)}{f(D)} \Leftrightarrow f(A)f(D) = f(E)f(F)$$

as desired. □

**Theorem 1.10** (CRL: circle + circle)

Let $ABCD$ and $BCEF$ be cyclic quadrilaterals with different circumcircles. Let $f(Z) = \pm\frac{BZ}{CZ}$ with the usual sign definition. Then

$$ADEF \text{ is cyclic} \Leftrightarrow f(A)f(D) = f(E)f(F)$$




*Proof.* By the Radical Axis Theorem, $ADEF$ is cyclic if and only if $AD$ and $EF$ concur on $BC$, which is true if and only if:

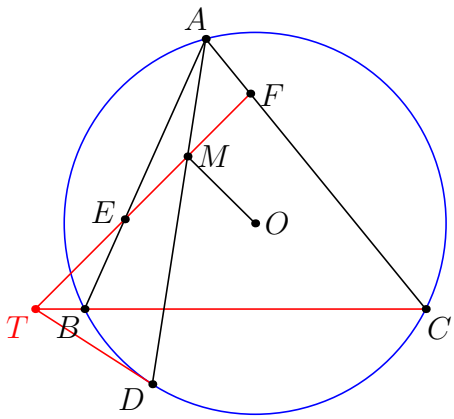
$$f(A)f(D) = f(E)f(F)$$

as desired. □

Sometimes, the choice of segment on which to define the cool-ratio function is not obvious, as it may happen that defining it on one segment makes the calculation of other points complicated. Let's see an example of this.

**Example 1.11** (Russian Handout[1])

Let $ABC$ be a triangle and $E, F$ be points on sides $AB$ and $AC$, respectively. Let $M$ be the projection of the circumcenter of $ABC$ onto $EF$. The line $AM$ intersects the circumcircle of triangle $ABC$ again at point $D$. Suppose that $EF$, $BC$, and the tangent to the circumcircle of $ABC$ at point $D$ are concurrent. Prove that $M$ is the midpoint of $EF$.




**Walkthrough.** If we try to define $f = f_{BC}$, we will quickly encounter difficulty calculating $f$ for other points. Furthermore, it is not obvious how to prove $M$ is the midpoint, as $f_{BC}$ doesn't extract useful information.

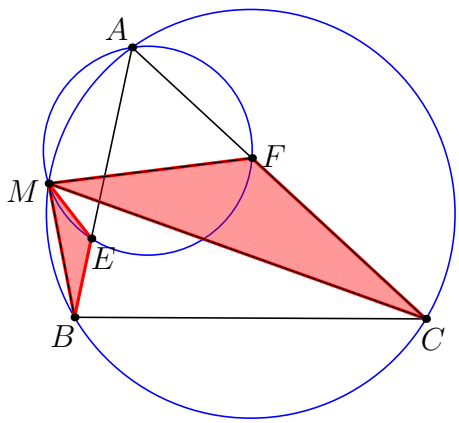
1. Consider the intersections of line $EF$ with $(ABC)$, let them be $X$ and $Y$.
2. Define $f_{XY} = f$.
3. Use Theorem 1.9 and Lemma 1.6.
4. Prove that $f(E) = \frac{1}{f(F)}$.

Finally, let's look at two important results.

**Theorem 1.12** (CRL: spiral similarity)

Let $ABC$ be a triangle and $E, F$ be points on lines $AB$ and $AC$ respectively. Let $M$ be the intersection of the circumcircles of $ABC$ and $AEF$. Then

$$|f_{BC}(M)| = \frac{BE}{CF}$$



*Proof.* With angle-chase, one easily concludes that

$$\triangle MBE \sim \triangle MCF.$$

Therefore:

$$|f_{BC}(M)| = \left|\pm\frac{BM}{CM}\right| = \frac{BE}{CF},$$

as desired. □

**Corollary 1.13** (CRL: inversion)

Let $A$ and $B$ be points on the line $XY$. Let $M$ be the midpoint of segment $XY$. The following statements are equivalent:

- $f_{XY}(A) = -f_{XY}(B)$ (or, equivalently, $(XY; AB) = -1$);
- $MA \cdot MB = MX^2$ (that is, points $A$ and $B$ are inverses with respect to the circle with diameter $XY$).

## §2 Practice problems

The number of [♣] corresponds to the approximate difficulty of the problems.

[2♣] **Required Problem 1** (Folklore)**.** Let $ABCD$ be a cyclic quadrilateral. Prove that $(AC; BD) = -1$ if and only if $AA, CC, BD$ are concurrent.

[2♣] **Required Problem 2** (Folklore)**.** Let $BAECFD$ be a cyclic hexagon. Suppose that $AF, DE, BC$ are concurrent and that $AE, DF, BC$ are concurrent. Prove that the quadrilaterals $BACD$ and $BECF$ are harmonic.

[2♣] **Required Problem 3** (Telegram)**.** Let $ABCDEF$ be a cyclic hexagon. Segments $BE$ and $CF$ intersect segment $AD$ at points $P$ and $Q$, respectively, such that point $P$ belongs to segment $AQ$. The circles $(BQE)$ and $(CPF)$ intersect line $AD$ again at points $R$ and $S$, respectively. Prove that $RA = SD$.

[2♣] **Problem 4.** Let $ABC$ be a triangle and $D$ be the intersection of the angle bisector of $\angle BAC$ with $BC$. Consider points $P, Q$ on the circumcircle of $ABC$ such that

$$\angle BPD = \angle DQC = 90^\circ.$$

Prove that the tangent at $A$, $BC$, and $PQ$ are concurrent.

[2♣] **Problem 5.** Given a triangle $ABC$. Let $M$ be the midpoint of side $AC$ and $P$ be the midpoint of segment $CM$. The circumcircle of $ABP$ intersects segment $BC$ at an interior point $Q$. Prove that $\angle ABM = \angle MQP$.

[2♣] **Problem 6** (Folklore)**.** Let $ABC$ be a triangle with circumcircle $\Omega$ and $X, Y$ points on line $BC$. Prove that $(XYA)$ is tangent to $\Omega$ if and only if $\angle BAX = \angle YAC$.

[2♣] **Problem 7.** Let $ABC$ be a triangle with circumcircle $\Omega$. Consider an arbitrary line $\ell$, which intersects $\Omega$ at distinct points $X$ and $Y$. Let $M$ be the Miquel point of lines $AB, BC, AC$, and $\ell$. Let $f = f_{XY}$. Prove that $f(M) = f(A) \cdot f(B) \cdot f(C)$.

[3♣] **Required Problem 8** (Folklore)**.** Let $\Omega$ be the Apollonian circle defined by points $X, Y$ and a constant $k$. Consider $A \in \Omega$ and let $O$ be the center of $\Omega$. Prove that

$$f_{XY}(O) = f_{XY}(A)^2.$$

[3♣] **Problem 9** (Telegram)**.** Let $ABCD$ be a cyclic quadrilateral with perpendicular diagonals and center $O$. The tangents to this circle at points $A$ and $C$, together with line $BD$, form a triangle $\Delta$. Prove that the circumcircles of $BOD$ and $\Delta$ are tangent.

[3♣] **Problem 10** (Kolmogorov Student Olympiads)**.** Given an isosceles trapezoid $ABCD$ ($AD \parallel BC$). Circle $\Omega$, with center at point $M$ on line $AD$, passes through points $B$ and $C$ and intersects the circumcircle of triangles $BAM$ and $BDM$ again at points $P$ and $Q$, respectively. Lines $PQ$ and $AD$ intersect at point $T$. Prove that line $TC$ is tangent to circle $\Omega$.

[3♣] **Problem 11** (Telegram)**.** Let $ABCD$ be a harmonic quadrilateral, $R$ the intersection point of its diagonals, and $L$ the foot of the angle bisector from $C$. The circle passing through $C$, $R$, and $L$ intersects $(ABCD)$ at point $Y$, and the circle passing through $A$, $R$, and $L$ intersects the same circle at point $X$.

Prove that the center of circle $XYR$ lies on line $BD$.

[3♣] **Problem 12** (2023 August)**.** In acute triangle $\triangle ABC$ with $AB < AC$, $AD$ is the angle bisector of $\angle BAC$ and $M$ is the midpoint of $BC$. Let $\Gamma$ be the circumcircle of $\triangle ABC$. $E, F$ are on $\Gamma$ such that $AE \perp DE$ and $AF \perp DA$. The circle passing through $A, E, M$ intersects $BC$ again at $G$. $FG$ intersects $\Gamma$ at $P$ and the tangent to $\Gamma$ at $E$ intersects line $BC$ at $Q$. Prove that $P, Q, E, D$ are concyclic.

[3♣] **Problem 13** (Telegram)**.** Given cyclic quadrilateral $ABCD$, lines $AD$ and $BC$ intersect at point $X$ and lines $AC$ and $BD$ at point $Y$. The angle bisectors of $\angle AXB$ and $\angle AYB$ intersect $AB$ at points $U$ and $V$. Prove that $C, D, U, V$ are concyclic.

[5♣] **Problem 14** (2024 RMM G2)**.** Let $ABC$ be an acute triangle with orthocenter $H$ and circumcircle $\Gamma$. Let $D$ be the point diametrically opposite to $A$ on $\Gamma$. The line passing through $H$, parallel to $BC$, intersects $AB$ and $AC$ at $X$ and $Y$, respectively. Line $AD$ intersects the circumcircle of triangle $DXY$ again at $S$. Let $T$ be the intersection of the tangent to $\Gamma$ at $A$ with $XY$. Prove that lines $DT$ and $HS$ intersect on $\Gamma$.

[5♣] **Problem 15** (3rd AGO P1, MathLuis)**.** Given a triangle $\triangle ABC$ with circumcenter $O$, let $\ell_B, \ell_C$ be lines passing through $A$ perpendicular to $AB, AC$, respectively. Suppose the reflections of $\ell_B, \ell_C$ across $BC$ intersect $OC, OB$ at $K, L$, respectively. Show that if the circumcircles of triangles $\triangle BOC$ and $\triangle KOL$ intersect again at a point $G$, then $\angle AGO = 90°$.

[5♣] **Problem 16** (2011 IMOSL G4)**.** Let $ABC$ be an acute triangle with circumcircle $\Omega$. Let $B_0$ be the midpoint of $AC$ and $C_0$ be the midpoint of $AB$. Let $D$ be the orthogonal projection of $A$ on $BC$, and let $G$ be the centroid of triangle $ABC$. Let $\omega$ be a circle passing through $B_0$ and $C_0$ that is tangent to circle $\Omega$ at a point $X \neq A$. Prove that points $D$, $G$, and $X$ are collinear.

[9♣] **Problem 17** (Kolmogorov Contest)**.** Let $ABCD$ be a tangential quadrilateral (it has an incircle). Lines $AB$ and $CD$ intersect at point $E$, and lines $AD$ and $BC$ intersect at point $F$. The external bisectors of angles $A$ and $C$ and line $EF$ form triangle $\Delta_1$. The external bisectors of angles $B$ and $D$ and line $EF$ form triangle $\Delta_2$. Prove that the circumcircles of triangles $\Delta_1$ and $\Delta_2$ are tangent.

[9♣] **Problem 18** (2024 IMOSL G7)**.** Let $ABC$ be a triangle with incenter $I$ such that $AB < AC < BC$. The second intersections of $AI$, $BI$, and $CI$ with the circumcircle of triangle $ABC$ are $M_A$, $M_B$, and $M_C$, respectively. Lines $AI$ and $BC$ intersect at $D$, and lines $BM_C$ and $CM_B$ intersect at $X$. Suppose the circumcircles of triangles $XM_BM_C$ and $XBC$ intersect again at $S \neq X$. Lines $BX$ and $CX$ intersect the circumcircle of triangle $SXM_A$ again at points $P \neq X$ and $Q \neq X$, respectively.

Prove that the circumcenter of triangle $SID$ lies on line $PQ$.

[9♣] **Problem 19** (3rd AGO P6)**.** Let $ABC$ be a triangle with circumcenter $O$. Let $\omega_B$ and $\omega_C$ be the $A$-excircles of triangles $AOB$ and $AOC$, respectively.

Prove that the circumcircle of the triangle formed by the external tangents to $\omega_B$, $\omega_C$ and line $BC$ is tangent to the circumcircle of $ABC$.

[9♣] **Problem 20** (Russian handout[1])**.** Let $ABC$ be a triangle, median $AM$ intersects the circumcircle of $ABC$ again at point $F$. Consider a parabola inscribed in triangle $ABC$ with focus $F$ touching lines $BC, AC$, and $AB$ at points $P, Q$, and $R$, respectively. Prove that points $P, Q, R$, and $M$ are concyclic.

[9♣] **Problem 21.** Let $\omega$ be a circle and $XY$ be a chord. Let $A, B, C, D$ be points on $\omega$ and $E, F$ be points on $XY$. Prove that $A, B, C, D, E, F$ lie on a conic if and only if:

$$f(A) \cdot f(B) \cdot f(C) \cdot f(D) = f(E) \cdot f(F)$$

## References

[1] Yu. E. Nagumanov, S. M. Kuznetsov, L. P. Gorbunov, A. A. Zaslavsky. *Poncelet invariants in light of Cool ratio lemma*. LKTG (Summer Conference of the Tournament of Towns), 2025.

[2] mira74. *Ratio Lemma*. Art of Problem Solving (AoPS), 2020. Available at: https://artofproblemsolving.com/community/q1h2357938p19166714.